Kunal Jha

Website | Google Scholar | Github | LinkedIn | Email

Education

University of Washington, Seattle, WA — June 2029

PhD Candidate in Computer Science and Engineering | Advisor: **Natasha Jaques** & **Max Kleiman-Weiner**

- Researching multi-agent systems, reinforcement learning, and computational cognitive science

Dartmouth College, Hanover, NH — June 2024

Bachelor of Arts, Major in Computer Science and Philosophy | Advisor: **Alberto Quattrini Li** & **Jeremy Manning**

- Coursework: Deep Learning, Multi-Robot Systems, Artificial Intelligence, Reinforcement Learning

Publications

- *Modeling Others' Minds as Code*
  (ICLR 2026, Best Paper NeurIPS 2025 LAW workshop)
  Kunal Jha, Aydan Yuenan Huang, Eric Ye, Natasha Jaques*, Max Kleiman-Weiner*
- *Cross-Environment Cooperation Enables Zero-shot Multi-agent Coordination*
  (Oral Presentation/Top 1% paper ICML 2025, Abstract CogSci 2025)
  Kunal Jha, Wilka Carvalho, Yancheng Liang, Simon S. Du, Max Kleiman-Weiner*, Natasha Jaques*
- *GOMA: Proactive Embodied Cooperative Communication via Goal-Oriented Mental Alignment*
  (IROS 2024)
  Lance Ying, Kunal Jha, Shivam Aarya, Joshua B. Tenenbaum, Antonio Torralba, Tianmin Shu
- *RIPL: Recursive Inference for Policy Learning*
  (Undergrad Thesis 2024)
  Kunal Jha, Jeremy Manning, Alberto Quattrini Li
- *Neural Amortized Inference for Nested Multi-agent Reasoning*
  (AAAI 2024, Oral Presentation @ AAAI Summer Symposium 2024)
  Kunal Jha, Tuan Anh Le, Chuanyang Jin, Yen-Ling Kuo, Joshua B. Tenenbaum, Tianmin Shu
- *Exploring high-order network dynamics in brains and stock markets*
  (Wetterhahn Symposium 2023)
  Kunal Jha, Daniel Carstensen, Ansh Patel, Jeremy Manning

Research Experience

Social Reinforcement Learning Lab, University of Washington, Seattle, WA — June 2024 - Present

Advised by Natasha Jaques and Max Kleiman-Weiner

- Researching how AI can coordinate with novel partners on novel problems through domain randomization and self-play. Built accessible tool for human-AI and human-human evaluation on arbitrary MARL benchmarks.
- Using program synthesis and LLMs to support scalable Theory of Mind reasoning in embodied domains

Reality and Robotics Lab, Dartmouth College, Hanover, NH — June 2022 - June 2024

Advised by Alberto Quattrini Li

- Conducting honors thesis on simultaneous probabilistic reasoning and heterogeneous policy learning. Leveraging graph neural networks and Bayes' rule to design an algorithm 2x faster than prior work.
- Created reward prediction model for stream-based active learning and state uncertainty estimation in Atari agents. Research was the capstone project for independent study on reinforcement learning.

Computational Cognitive Science Lab, MIT, Cambridge, MA — March 2022 - June 2024

Research Assistant to Joshua Tenenbaum

- Utilized Deep Learning, Importance Sampling, and Monte Carlo methods to emulate human nested social inference capabilities in autonomous-driving and construction domains.
- Collaborated with researchers at **Google** to innovate online particle inference and amortized tree search algorithm for multi-agent interaction that is more accurate and 16x faster than existing benchmarks.
- Incorporating MCTS and a divergence-based approach for effective communication by LLM-backed agents looking to collaborate with humans on the VirtualHome and Overcooked benchmarks.

Contextual Dynamics Lab, Dartmouth College, Hanover, NH — January 2021 - June 2024

Research Assistant to Jeremy Manning

- Applied correlation-based kernel filtering process for next-time prediction of fMRI and stock data that is robust to lower-order network activity spikes, resulting in error reductions exceeding 13%.
- Adopted scrum methodology to manage 6+ undergraduate students on a fintech project, meeting with the P.I and hosting weekly code reviews + pair programming sessions to accomplish sprints and coordinate upcoming project deliverables.
- Created an experiment to understand the neuronal processes behind conceptualization during and after educational lectures. Implemented topic model to quantify patterns between stimuli and treatment questions. Extracted sentence embeddings from video transcripts to semantically map fMRI trial data.

Teaching

- Deep Learning Teaching assistant (35 students) — spr’ 2024
- Foundations of Applied Computer Science Teaching assistant (70+ students) — wint’ 2023
- Problem Solving via Object Oriented Programming Teaching assistant (80+ students) — aut’ 2022

Industry Experience

Google, Zurich, Switzerland — January 2026 - May 2026
Student Researcher
- Working with the Paradigms of Intelligence team on unifying game theory and ALife through thermodynamics

Amazon Web Services, Arlington, VA — June 2024 - September 2024
Software Development Engineer Intern
- Automated the differential testing pipeline for the Verified Permissions team by developing AWS Lambda and Java wrappers to execute and analyze Rust code of Cedar policy management language.

Amazon Web Services, Arlington, VA — June 2023 - September 2023
Software Development Engineer Intern
- Built end-to-end pipeline for alerting users about server-side health events impairing their workflows and recommending solutions to their issues, simultaneously developing back-end API and front-end widget.
- Adapted sampling-cost solutions to the optimal stopping problem to efficiently provide recommendations while increasing case deflections by 0.5% in simulated model

National Aeronautics and Space Administration (NASA), Cape Canaveral, FL — December 2020 - April 2021
Financial Data Science Intern
- Constructed the MUREP program’s first holistic financial dataset by parsing 20 years of textual data, then implemented a Deep Learning agent that recommended a $35 million research budget redistribution.
- Performed sentiment analysis on grant recipient reports and used results to retrain the recommendation model (to proxy for MUREP’s qualitative objectives), improving struggling institution success rates by 14%.

Awards

- Cooperative AI Early Career Grant Recipient — 2025
- Sony Focused Research Award Recipient — 2025
- CSERF Fellowship — 2024
- ARCS Scholar — 2024
- High Honors in Computer Science at Dartmouth College — 2024
- Citation for Academic Excellence in Philosophy and Computers — spr’ 2024
- Dean’s Honor List — 2023
- Citation for Academic Excellence in Deep Learning, Robustness, and Generalization — spr’ 2023
- Citation for Academic Excellence in Computer Vision — wint’ 2023
- Citation for Academic Excellence in Multi-robot Systems — aut’ 2022
- Lovelace Computing Scholar (for novel approaches to computational research) - $1200 — spr’ 2022
- SELF Grant Recipient (for experiential learning and interuniversity research) - $5000 — spr’ 2022

- 5x URAD Scholar (for promising research amongst sophomores and juniors) - $6000 — 2021, 2022
- Great Issues Scholar (for first-years looking to engage with international issues) — 2021
- National Merit Scholar Finalist — 2020

Skills

Machine social intelligence, Human-robot interaction, **artificial intelligence**, embodied cognition, computational cognitive science, machine learning, **reinforcement learning**, deep learning, **probabilistic programming**, Bayesian modeling and inference, Monte Carlo methods, Python, Pytorch, Jax, Java, C, fMRI trained, project management, public speaking, outreach, **mentorship**

Languages

- English (Native)
- Hindi (Fluent)
- German (Conversational)